MATH1231 Mathematics 1B – Algebra

Section 1: Introduction to Vector Spaces

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Investigation

Exercise 1.

1. Let $\overrightarrow{\mathbf{a}} = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 2 \end{pmatrix}$ and $\overrightarrow{\mathbf{b}} = \begin{pmatrix} 3 \\ 1 \\ -2 \\ -1 \end{pmatrix}$ be vectors in $\mathbb{R}^4$.
   Evaluate $\overrightarrow{\mathbf{c}} = 3\overrightarrow{\mathbf{a}} + 2\overrightarrow{\mathbf{b}}$.

2. Let $A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix}$ be matrices in $M_{2,2}$.
   Evaluate $C = 3A + 2B$.

3. Let $p(x) = 1 - x + 2x^3$ and $q(x) = 3 + x - 2x^2 - x^3$ be polynomials.
   Evaluate $r = 3p + 2q$.

Question. What do you notice about these problems?

Investigation

Exercise 2.

1. Simplify $2(\overrightarrow{\mathbf{a}} + 4\overrightarrow{\mathbf{b}}) + \overrightarrow{\mathbf{a}}$, where $\overrightarrow{\mathbf{a}}$ and $\overrightarrow{\mathbf{b}}$ are vectors in $\mathbb{R}^5$.

See if you can give the answer to the next question with *absolutely no working!*

2. Simplify $2(A + 4B) + A$, where A, B are matrices in $M_{3,7}$.

Vector spaces

Since all of the examples on the previous page – and many more! – are "really the same", we shall study them together instead of individually.

- This will be more efficient, as we won't have to do the same work over and over again.
- Studying what all of these systems have in common will enable us to gloss over superficial differences and concentrate on gaining a better understanding of what is *really* important in all of them.

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The term used for all of these systems when we study them simultaneously is **vector space**.

What makes up a vector space?

What do we actually need in order to perform the kind of calculations we did on page 2?

- The "objects" that we want to calculate with. These will be referred to as vectors. Note that from now on, vectors need not be things like $(1, -1, 0, 2)$, but may also be matrices, polynomials, functions and many other things.

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- We need to know how to add two vectors.
- We need a set of numbers. These numbers should form a field – that is, we must be able to add, subtract, multiply and divide any two numbers (except for division by zero) in accordance with "sensible" rules. In vector space theory the numbers are commonly referred to as scalars. For this course scalars will usually be real numbers, occasionally complex numbers. There are areas of mathematics where other fields are also important.

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Note that although we may in some cases know how to multiply a vector by a vector (for example...?), this does not form part of vector space theory – at least, not in first year.

Field axioms (MATH1131 revision)

Recall from MATH1131 that $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are fields. More generally, a set $\mathbb{F}$ is a **field** iff it satisfies the the following axioms:

- Closure under addition: $x + y \in \mathbb{F}$ for all $x, y \in \mathbb{F}$.
- Additive associativity: $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}$.
- Additive commutativity: $x + y = y + x$ for all $x, y \in \mathbb{F}$.
- Additive identity: There exists $0 \in \mathbb{F}$ such that $x + 0 = x$ for all $x \in \mathbb{F}$.
- Additive inverse: For all $x \in \mathbb{F}$ there is $-x \in \mathbb{F}$ such that $x + (-x) = 0$.
- Closure under multiplication: $xy \in \mathbb{F}$ for all $x, y \in \mathbb{F}$.
- Multiplicative associativity: $(xy)z = x(yz)$ for all $x, y, z \in \mathbb{F}$.
- Multiplicative commutativity: $xy = yx$ for all $x, y \in \mathbb{F}$.
- Multiplicative identity: There exists $1 \in \mathbb{F}$ such that $1x = x$ for all $x \in \mathbb{F}$.
- Multiplicative inverse: For all $x \in \mathbb{F}$ where $x \neq 0$, there exists $x^{-1} \in \mathbb{F}$ such that $xx^{-1} = 1$.
- Distributivity: $x(y + z) = xy + xz$ for all $x, y, z \in \mathbb{F}$.

What rules should addition of vectors and multiplication by scalars obey?

To find out part of the answer, let's do exercise 5 from page 2 again, one step at a time, noting what properties we use:

$$
\begin{aligned}
2(\overrightarrow{\mathbf{a}} + 4\overrightarrow{\mathbf{b}}) + \overrightarrow{\mathbf{a}} &\overset{(1)}{=} (2\overrightarrow{\mathbf{a}} + 2(4\overrightarrow{\mathbf{b}})) + \overrightarrow{\mathbf{a}} && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (1)\\
&\overset{(2)}{=} (2\overrightarrow{\mathbf{a}} + (2\times 4)\overrightarrow{\mathbf{b}}) + \overrightarrow{\mathbf{a}} && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (2)\\
&\overset{(3)}{=} (2\overrightarrow{\mathbf{a}} + 8\overrightarrow{\mathbf{b}}) + \overrightarrow{\mathbf{a}} && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (3)\\
&\overset{(4)}{=} \overrightarrow{\mathbf{a}} + (2\overrightarrow{\mathbf{a}} + 8\overrightarrow{\mathbf{b}}) && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (4)\\
&\overset{(5)}{=} (\overrightarrow{\mathbf{a}} + 2\overrightarrow{\mathbf{a}}) + 8\overrightarrow{\mathbf{b}} && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (5)\\
&\overset{(6)}{=} (1\overrightarrow{\mathbf{a}} + 2\overrightarrow{\mathbf{a}}) + 8\overrightarrow{\mathbf{b}} && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (6)\\
&\overset{(7)}{=} (1+2)\overrightarrow{\mathbf{a}} + 8\overrightarrow{\mathbf{b}} && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (7)\\
&\overset{(8)}{=} 3\overrightarrow{\mathbf{a}} + 8\overrightarrow{\mathbf{b}}. && \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & (8)
\end{aligned}
$$

Zero, negatives

The simplification on the previous page makes use of six properties which we would want to be true if addition and multiplication are to be "sensible". There are some more which were not used in the previous example.

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Exercise 4. What can you deduce from the equation

$$\overrightarrow{\mathbf{a}} + \overrightarrow{\mathbf{c}} = \overrightarrow{\mathbf{b}} + \overrightarrow{\mathbf{c}} \ ?$$

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Exercise 5. What can you deduce from the equation

$$\overrightarrow{\mathbf{a}} + \overrightarrow{\mathbf{c}} = \overrightarrow{\mathbf{b}} + \overrightarrow{\mathbf{c}} \ ?$$

Solution. We would like to subtract $\overrightarrow{\mathbf{c}}$ from both sides of the equation, that is, add the negative of $\overrightarrow{\mathbf{c}}$. Thus

$$\overrightarrow{\mathbf{a}} + \overrightarrow{\mathbf{c}} = \overrightarrow{\mathbf{b}} + \overrightarrow{\mathbf{c}}$$

$$\Rightarrow \quad (\overrightarrow{\mathbf{a}} + \overrightarrow{\mathbf{c}}) + (-\overrightarrow{\mathbf{c}}) = (\overrightarrow{\mathbf{b}} + \overrightarrow{\mathbf{c}}) + (-\overrightarrow{\mathbf{c}}) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \tag{9}$$

$$\Rightarrow \quad \overrightarrow{\mathbf{a}} + (\overrightarrow{\mathbf{c}} + (-\overrightarrow{\mathbf{c}})) = \overrightarrow{\mathbf{b}} + (\overrightarrow{\mathbf{c}} + (-\overrightarrow{\mathbf{c}})) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \tag{10}$$

$$\Rightarrow \quad \overrightarrow{\mathbf{a}} + \overrightarrow{\mathbf{0}} = \overrightarrow{\mathbf{b}} + \overrightarrow{\mathbf{0}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \tag{11}$$

$$\Rightarrow \quad \overrightarrow{\mathbf{a}} = \overrightarrow{\mathbf{b}} \ . \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \tag{12}$$

Closure

There are two properties that we haven't discussed yet, although arguably they are the most important of all.

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Well, if we start with vectors in $\mathbb{R}^3$ then the answer should be . . . a vector in $\mathbb{R}^3$,

if we start with 2×5 matrices then the answer should be a . . .

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We need the **closure laws**.

- If $\overrightarrow{\mathbf{u}}$ and $\overrightarrow{\mathbf{v}}$ are vectors, then $\overrightarrow{\mathbf{u}} + \overrightarrow{\mathbf{v}}$ is a vector "of the same kind" – that is, from the same vector space.
- If $\overrightarrow{\mathbf{v}}$ is a vector and λ a scalar, then $\lambda\overrightarrow{\mathbf{v}}$ is a vector "of the same kind".

These might sound totally obvious, but you have studied operations where the result produced from two objects is an object "of a different kind": for example...?

Vector space: the definition

Definition. A vector space over a field $\mathbb{F}$ is a non–empty set V of vectors, together with an operation of vector addition and an operation of multiplication by scalars from $\mathbb{F}$, for which the following properties hold.

1. Closure under (vector) addition. If $\overrightarrow{\mathbf{u}}, \overrightarrow{\mathbf{v}}$ are in V, then $\overrightarrow{\mathbf{u}} + \overrightarrow{\mathbf{v}}$ is in V.
2. Associativity of (vector) addition. For all $\overrightarrow{\mathbf{u}}, \overrightarrow{\mathbf{v}}, \overrightarrow{\mathbf{w}}$ in V we have $(\overrightarrow{\mathbf{u}} + \overrightarrow{\mathbf{v}}) + \overrightarrow{\mathbf{w}} = \overrightarrow{\mathbf{u}} + (\overrightarrow{\mathbf{v}} + \overrightarrow{\mathbf{w}})$.
3. Commutativity of (vector) addition. For all $\overrightarrow{\mathbf{u}}, \overrightarrow{\mathbf{v}}$ in V we have $\overrightarrow{\mathbf{u}} + \overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{v}} + \overrightarrow{\mathbf{u}}$.
4. Zero vector. There exists an element $\overrightarrow{\mathbf{0}}$ of V such that for all $\overrightarrow{\mathbf{v}} \in V$ we have $\overrightarrow{\mathbf{v}} + \overrightarrow{\mathbf{0}} = \overrightarrow{\mathbf{v}}$.
5. Negatives. For each element $\overrightarrow{\mathbf{v}}$ of V there is an element $\overrightarrow{\mathbf{w}}$ of V such that $\overrightarrow{\mathbf{v}} + \overrightarrow{\mathbf{w}} = \overrightarrow{\mathbf{0}}$. (We usually denote this element by $-\overrightarrow{\mathbf{v}}$.)
6. Closure under scalar multiplication. If λ is in $\mathbb{F}$ and $\overrightarrow{\mathbf{v}}$ is in V, then $\lambda\overrightarrow{\mathbf{v}}$ is in V.
7. "Associativity" of multiplication. For all $\lambda, \mu \in \mathbb{F}$ and all $\overrightarrow{\mathbf{v}} \in V$ we have $\lambda(\mu\overrightarrow{\mathbf{v}}) = (\lambda\mu)\overrightarrow{\mathbf{v}}$.
8. Multiplication by (the scalar) 1. For each $\overrightarrow{\mathbf{v}} \in V$ we have $1\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{v}}$.
9. Scalar distributive law. For all λ, μ in $\mathbb{F}$ and all $\overrightarrow{\mathbf{v}}$ in V we have $(\lambda + \mu)\overrightarrow{\mathbf{v}} = \lambda\overrightarrow{\mathbf{v}} + \mu\overrightarrow{\mathbf{v}}$.
10. Vector distributive law. For all λ in $\mathbb{F}$ and all $\overrightarrow{\mathbf{u}}, \overrightarrow{\mathbf{v}}$ in V we have $\lambda(\overrightarrow{\mathbf{u}} + \overrightarrow{\mathbf{v}}) = \lambda\overrightarrow{\mathbf{u}} + \lambda\overrightarrow{\mathbf{v}}$.

Helpful ways to think about vector spaces

The more you become involved with abstract mathematical concepts, the more you will find it helpful to develop a "dual perspective" on the things you study.

- You need to know precisely what a vector space is – that is, you need to know the ten axioms. We want to prove things about vector spaces, and to be certain that the results we obtain are reliable. This can only be done on the basis of an absolutely precise definition of the concept.

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- However, it is very difficult from the definition alone to acquire a good intuitive understanding – a "gut feeling" – for what is meant by a vector space. You need to supplement the definition by a good "rough idea" of what a vector space is. My own rough idea is this: *a vector space is "something where we know how to add objects and multiply objects by numbers, and where the addition and multiplication satisfy 'sensible' laws"*.

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It may help to visualise things in terms of the most familiar vector spaces, such as $\mathbb{R}^2$ and $\mathbb{R}^3$, as long as you **remember** that we may actually be working with vectors which appear very different from these.
It may also help to realise that none of the vector space axioms are really new to you! – they are all algebraic rules which you know very well for vectors, matrices, polynomials and so forth.

Vector space or not vector space?

TIP!

- When you add two elements of the set, do you get an element of that set / stay inside the set?
- When you multiply an element of the set by a scalar, do you stay inside the set?
- Is there a zero element in the set?

■ Students use https://kahoot.it/ or the Kahoot app on their phone.

■ The lecturer starts the game using:
https://create.kahoot.it/share/vector-space-or-not/dcfa83d2-45fc-4e38-9249-b55653b47dc2
or
Kahoot: Is this a vector space?

Examples of sets which are vector spaces

We have mentioned many examples of vector spaces already.

1. The vector spaces $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^4$ and in general $\mathbb{R}^n$. These are just "ordinary" vectors in the sense of Chapter 1 from MATH1131. They can be thought of either as n–tuples of real numbers, or as geometric vectors: in the latter case we can define addition and scalar multiplication by means of diagrams, and we can then use diagrams to prove that the vector space axioms hold.

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4. The set of all polynomials is a vector space.

Examples of sets which are NOT vector spaces

In order to properly understand any mathematical concept, it is important to see why some examples work, **and** *to see why other examples do not work.*

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3. The set of vectors in $\mathbb{R}^2$ with length less than 5 is not a vector space.

4. The set of vectors in $\mathbb{R}^2$ which point into the first quadrant is not a vector space.

5. The set of polynomials of degree 3 is not a vector space.

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3. Vectors in $\mathbb{R}^n$ do not form a vector space over $\mathbb{C}$.